Spyder

/Users/frankie/Dropbox/Harper College/CIS 2026/M8/M8P1.py

M6P4.py | M6P5.py | M7P1.py | M7P2.py | M7P3.py | M7P4.py | M7P5.py | M8P1.py

```python
    Description:
        Allow the user to enter a principal amount and interest rate. Repeatedly (need a loop to c
    the program execution) compute the annual interest (principle x rate). Compute ending balance
    to be principal (beginning balance + interest). Display year, beginning balance and ending
    balance for each of the first 5 years. Display the accumulated interest for the 5 years. Note:
    new balance by year (this will be the principle for the following year. Format the output to l
    like the example below.
    Example:
    Enter principal amount: 10000.00
    Enter interest rate: 0.10
    Formatted output
    Year Beginning Ending
    Balance Balance
    1 $10,000.00 $11,000.00
    2 $11,000.00 $12,100.00
    3 $12,100.00 $13,310.00
    4 $13,310.00 $14,641.00
    5 $14,641.00 $16,105.00
    Total interest earned: $6,156.00
    """

    principleAmount = 0.0
    interestRate = 0.0
    annualInterest = 0.0
    accumlatedInterest = 0.0
    NO_OF_YEAR = 5

    # User Enter the value
    principleAmount = float(input("Enter a principal amount: "))
    interestRate = float(input("Enter a interest rate: "))
    print("Year:   Beginning     Ending")
    print("          Balance    Balance")
    print("".ljust(28,"-"))

    for i in range(NO_OF_YEAR):
        # Calculate annual interest and print out
        annualInterest = principleAmount * interestRate
        print(f"{i+1:4.0f}   {principleAmount:10,.2f} {principleAmount + annualInterest:10,.2f}")

        # Calculate the accumlated Interest and update the principle amount for next year
        accumlatedInterest += annualInterest
        principleAmount += annualInterest

    # Print out the accumlated interest
    print(f"Total interest earned: ${accumlatedInterest:10,.2f}")
```

...ropbox/Harper College/CIS 2026/M8

Name	Date Modified
norton	3/5/26 7:18 PM
M8P1.py	3/13/26 12:38 AM
M8P2.py	3/12/26 11:34 PM
M8P3.py	3/13/26 12:07 AM
M8P3Data.txt	3/13/26 12:03 AM
M8P4.py	3/13/26 12:24 AM
M8P4Data.txt	3/13/26 12:09 AM

Help | Variable Explorer | Debugger | Profiler | Plots | Files

Console 1/A

```
In [64]: %runfile '/Users/frankie/Dropbox/Harper College/CIS 2026/M8/
M8P1.py' --wdir
Enter a principal amount: 10000
Enter a interest rate: 0.1
Year:   Beginning     Ending
          Balance    Balance
----------------------------
   1    10,000.00  11,000.00
   2    11,000.00  12,100.00
   3    12,100.00  13,310.00
   4    13,310.00  14,641.00
   5    14,641.00  16,105.10
Total interest earned: $  6,105.10

In [65]:
```

IPython Console | History

Update available | Inline | Conda: spyder-runtime (Python 3.12.11) | LSP: Python | Line 43, Col 46 | UTF-8 | CRLF | RW | Mem 49%